PRINTSMART : A Web-Based E-Commerce and Order Management System for Printagram Printing Services

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Abstract—

I. Introduction

Printing services in the Philippines remain in high demand despite evolving technology. In the Asia-Pacific region, the Philippines is recorded as the largest consumer of printing presses, with approximately 1.1 million units and a market value of around $2.1 billion. The industry demonstrates annual growth rates of +53.6% in units and +40.0% in value between 2013 and 2024 (IndexBox, 2024). These figures indicate that the country's printing industry remains a vital component of business operations and consumer services. Within this landscape, Printagram Printing Services utilizes Facebook for marketing and client communication, benefiting from the platform's low-cost exposure to reach this growing market.

However, managing orders strictly through social media is disorganized, resulting in lost details, repeated inquiries, delays, and errors. Printagram Printing Services currently lacks a clear system to track and manage orders efficiently, leading to operational inefficiencies and reduced customer satisfaction. This issue is supported by Cho, Sutton, and Taskin (2023), who find that although social media usage by internet retailers is positively associated with service performance in online procurement, it does not significantly enhance order-fulfillment quality. This finding reveals a critical gap in information symmetry between front-end (order acquisition) and back-end (order fulfillment) operations, underscoring the need for a more organized and automated system.

To address these challenges, this study focuses on the development of a web-based e-commerce and order management system designed to improve order management, reduce mistakes, and enhance the customer experience. The system features a user account management module, a product and service catalog, an e-commerce shopping cart, checkout and payment gateway integration, and automated email notifications for customers. On the administrative side, the system includes secure login access, a central dashboard, and order management tools for efficient monitoring. Ultimately, this system aims to improve service accuracy, enhance administrative efficiency, and promote a more reliable and convenient ordering experience for both clients and company personnel.

A. Theoretical Framework

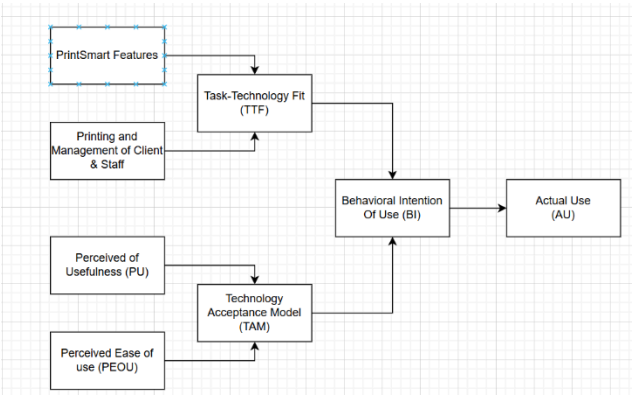


Figure 1. Theoretical Framework of the Study

This study is anchored on the integration of the **Technology Acceptance Model (TAM)** and the **Task-Technology Fit (TTF)** model. The combination of these theories illustrates that the success of the PrintSmart system depends not only on its ease of use but also on its functional alignment with the specific operational needs of the business. The framework posits that a strong fit between the technology and the user's tasks (TTF) enhances perceptions of usefulness and ease of use (TAM), which ultimately drives the intention to adopt the system.

Technology Acceptance Model (TAM) Proposed by Davis (1989), the Technology Acceptance Model asserts that system adoption is primarily driven by two cognitive factors: **Perceived Usefulness (PU)** and **Perceived Ease of Use (PEOU)**. In the context of this study, PU refers to the belief that PrintSmart will enhance performance by replacing disorganized manual logs with a centralized

dashboard for the admin and providing transparent tracking for the client. PEOU refers to the expectation that the system will be effortless to navigate. By offering a structured shopping cart and intuitive interface, the system is designed to be more efficient than the current method of manually sending order details via chat, thereby increasing the user's behavioral intention to use the platform.

Task-Technology Fit (TTF) Model Complementing TAM, the Task-Technology Fit model (Goodhue & Thompson, 1995) argues that technology will only improve performance if its capabilities align with the tasks users must perform. This model evaluates the correspondence between **Task Characteristics** (such as ordering, paying, and tracking) and **Technology Characteristics** (such as automation, receipt generation, and notifications). The study utilizes TTF to demonstrate that PrintSmart's specialized features offer a superior "fit" for printing service operations compared to the current Facebook Messenger-based process, ensuring that the system effectively supports the actual workflow of both staff and customers.

B. Conceptual Framework

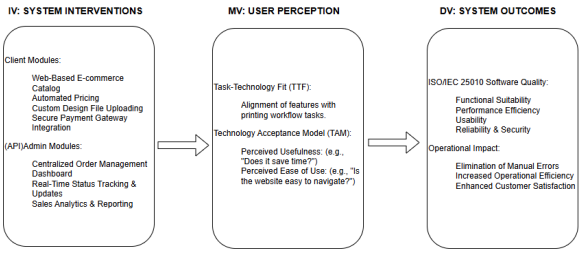


Figure 2. Conceptual Framework of the Study

C. Statement of the Problem

Specifically, this study seeks to answer the following questions:

1. What are the operational inefficiencies encountered by Printagram Printing Services in their current manual, Facebook-based process regarding order tracking, file management, and payment verification, and how does this limit their ability to reach a wider market?
2. What features must be developed for the PrintSmart Client-Side (e.g., catalog, cart, automated pricing) and Admin-Side (e.g., dashboard, monitoring) to resolve these issues and facilitate online transactions?
3. What is the software quality level of PrintSmart as evaluated by users based on ISO/IEC 25010, specifically in terms of Functional Suitability, Usability, Security, Reliability, and Performance Efficiency?

D. Objective of the Study

Specifically this study will aim to:

1. To replace Printagram's current manual and social media-based operations with a unified web platform that addresses inefficiencies in order tracking, file management, and payment verification, thereby enabling the business to reach a wider market.
2. To design and develop the PrintSmart system with the following key modules:

   Client-Side: A user-friendly interface featuring an e-commerce catalog, shopping cart, file upload capability, and secure payment gateway to facilitate seamless online transactions.

   Admin-Side: A centralized dashboard for real-time order monitoring, automated email notifications for status updates, and a structured workflow to minimize operational errors.

   3. To evaluate the quality of the developed PrintSmart system as assessed by client and admin users based on ISO/IEC 25010 software quality standards, specifically focusing on Functional Suitability, Usability, Security, Reliability, and Performance Efficiency.

A. Scope and Limitation of the Study

The study focuses on the design and development of PrintSmart, a web-based e-commerce and order management system aimed at digitalizing Printagram Printing Services' operations. The system features an Admin Dashboard for managing products, inventory, and order statuses, alongside a Client Interface that facilitates account registration, service browsing, custom file uploads, and secure online payments via API integration (GCash/Bank cards). However, the study is limited to a web-based platform accessible via browsers; it does not include the development of standalone Android or iOS mobile applications. Furthermore, the system excludes real-time third-party logistics tracking (e.g., Lalamove API) due to external agreement constraints; delivery updates will instead be managed manually by administrators.

B. Significance of the Study

This study contributes to the achievement of the United Nations Sustainable Development Goals (SDGs) 2030, specifically supporting SDG 4 (Quality Education), SDG 7 (Affordable and Clean Energy), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 17 (Partnerships for the Goals). Through the development and implementation of the Web-based E-commerce and Order Management System,

the study promotes digital innovation, operational efficiency, and sustainable business practices within the printing service industry.

For the business owner, the system provides a centralized digital platform that consolidates order tracking, inventory management, and sales reporting. This enables accurate monitoring of business performance, supports data-driven decision-making, and strengthens management control. By reducing reliance on manual logbooks, the system improves transparency, accountability, and long-term operational sustainability, directly contributing to SDG 8 and SDG 9. Simultaneously, the staff gains from the automation of front-desk operations such as order intake, file management, and receipt generation. The system minimizes clerical errors, reduces workload, and improves service efficiency, allowing staff to focus on production quality. This enhancement of workplace efficiency and skill utilization aligns with SDG 8, which promotes productive employment and decent work conditions.

From a customer and environmental perspective, the system offers distinct advantages. Clients benefit from a more convenient and accessible service experience through online ordering, real-time status tracking, and digital file submissions. These features promote consistent participation and digital literacy, supporting SDG 4 by encouraging the adoption of technology-enabled services for academic and personal needs. Environmentally, the system reduces the need for paper-based order slips and physical records, indirectly lowering resource consumption and energy waste associated with traditional administrative processes. This supports SDG 7 by promoting the efficient use of digital systems and sustainable operational practices.

Finally, for future researchers, this study serves as a reference for the design and implementation of web-enabled transaction and management systems. It provides insights into system architecture, technology integration, and process automation that can support collaborative research, system enhancement, and cross-sector innovation, aligning with SDG 17. Overall, the study highlights how digital transformation in small-to-medium enterprises can contribute to sustainable development by improving operational efficiency, fostering innovation, enhancing user experience, and supporting inclusive and technology-driven growth.

II. LITERATURE REVIEW

The transition from manual social commerce to dedicated e-commerce platforms is a critical evolution for modern business operations. Contemporary literature suggests that while social commerce offers low-cost exposure, it often lacks the structural efficiency required for scaling operations. Understanding the intricacies of online commerce has become essential for success in today's digital landscape **[1]**, with trends indicating a shift from market-centric to user-centric commerce that optimizes user engagement **[2]**. In the Philippine context, e-commerce is defined not just by online selling, but by the technological infrastructure that facilitates inventory management, payments, and customer service **[3]**. Studies indicate that strategic web engineering solutions can accelerate this adoption, enhancing customer engagement and supporting economic growth **[4]**. Furthermore, the feasibility of customizing these platforms for smaller, community-based retail environments has been demonstrated, proving that open-source solutions can effectively facilitate interaction between vendors and local customers **[5]**.

In the specific domain of custom printing services, web-based order management systems have been proven to automate transactions and enable real-time information flow. Research utilizing the Technology Acceptance Model (TAM) highlights that user acceptance is heavily influenced by perceived usefulness and ease of use, with web-based systems increasing service reliability **[6]**. Local studies reinforce this, emphasizing that centralized data management reduces manual workload while improving the speed of order handling **[7]**. Prototypes developed for printing shops have successfully streamlined order processing, reduced errors, and enhanced the overall user experience **[8]**, **[9]**.

A critical component of these systems is the integration of secure payment gateways. The security of financial transactions is a primary concern for consumers, and providing confidence that data is shielded from fraud is essential for building trust **[10]**. While the complexity of API integration poses challenges, selecting a gateway with a manageable integration process is vital to prevent transaction failures **[11]**. In the Philippines, the digital payments landscape is dominated by e-wallets like GCash, which accounts for a significant market share and is widely trusted for everyday transactions **[12]**, **[13]**. Empirical studies grounded in TAM suggest that perceived security and simplicity are key factors in building trust in these digital payment platforms **[14]**, and local consumers, while appreciative of convenience, remain highly sensitive to security issues such as transaction reversals **[15]**.

Finally, the design of these platforms must be user-centric to ensure adoption. Systems tailored to the specific needs and abilities of end-users significantly improve usability and satisfaction **[16]**. Research confirms that e-commerce platforms with clear workflows, aesthetic simplicity, and responsive designs minimize cognitive load and increase adoption rates **[17]**, **[18]**. Among Filipino micro-enterprises, simple navigation menus and localized content have been found to significantly influence customer satisfaction **[19]**. Furthermore, the inclusion of specific features such as organized product catalogs, dynamic pricing, and file upload capabilities aligns with the Task-Technology Fit (TTF) model, ensuring that the technology effectively meets the users' ordering tasks **[20]**, **[21]**.

III. METHODS

This chapter describes the research methods and procedures used in the development and evaluation of PrintSmart, a web-based e-commerce and order management system for Printagram Printing Services. It outlines the research approach, system development process, tools and technologies used, and the methods applied to assess the system's effectiveness in addressing the specific inefficiencies in manual order tracking, file management, and payment verification identified in the study's objectives.

A. Research Design

This study utilizes a combined developmental and descriptive mixed methods research design to guide the creation and evaluation of the system. The developmental design is utilized for the software engineering process required to construct the technical solution, specifically addressing the first and second objectives of the study which are to replace the manual Facebook-based operations and develop the Client-Side and Admin-Side modules. On the other hand, the descriptive design is used to evaluate the system's quality by gathering quantitative data. This phase systematically assesses the software quality level of PrintSmart based on ISO/IEC 25010 standards, specifically measuring Functional Suitability, Usability, Security, Reliability, and Performance Efficiency as outlined in the third statement of the problem.

B. Data Generation Procedure

Data for this study will be generated by collecting information related to the operational inefficiencies encountered by Printagram Printing Services. To address the first statement of the problem, existing manual records, including Facebook Messenger chat logs and manual order logbooks, will be reviewed to identify specific bottlenecks such as disorganized tracking, lost details, and payment verification errors. Semi-structured interviews will be conducted with the Printagram owner and key staff to document these pain points and define the functional requirements for the new system. Following the development of the Client-Side e-commerce interface and Admin-Side dashboard, the system will be deployed for user testing. During this phase, survey questionnaires based on ISO/IEC 25010 will be distributed to both administrative staff and client users to generate quantitative data regarding the system's quality. Additionally, qualitative data will be generated through open-ended feedback mechanisms to capture user experiences regarding task-system alignment. All collected data will be validated through pilot testing of the instruments and expert review to ensure data integrity.

C. Data Case Analysis

Collected data will be rigorously processed to ensure accuracy before analysis. Invalid and incomplete responses will be excluded to maintain the reliability of the study. The quantitative data derived from the survey responses will be encoded and analyzed using statistical software. These responses, based on a 5-point Likert scale, will be processed using descriptive statistics to calculate the Weighted Mean and Standard Deviation. This analysis will determine the level of acceptability for the system's Functional Suitability, Usability, Security, Reliability, and Performance Efficiency, directly answering the third research question. Simultaneously, qualitative data from interviews and feedback forms will be grouped and analyzed using Thematic Analysis. This process involves categorizing comments into themes such as "Usability Issues" or "Operational Improvements" to evaluate how well the system features are, such as the e-commerce catalog and automated pricing, to resolve the specific operational issues identified in the study. The quantitative and qualitative findings will be triangulated to draw comprehensive conclusions.

D. Statistical Treatment of Data

Locale of the Study

The study will be conducted at Printagram Printing Services, where the current manual and Facebook-based ordering process is implemented and where the new PrintSmart system will be deployed for testing.

Population of the Study

The population will consist of two distinct user groups: the administrative staff of Printagram Printing Services who will operate the Admin-Side dashboard, and the active and prospective client users who will utilize the Client-Side e-commerce interface.

Sample and Sampling Technique

The sample will consist of the entire administrative workforce of five personnel (Total Enumeration) to ensure all operational perspectives are captured . For the client users, a sample of approximately 30 to 40 individuals will be selected using Purposive Sampling. This technique ensures that the respondents chosen are familiar with the manual workflow or represent the target market, making them qualified to evaluate the improvements in order tracking and online transactions.

Statistical Treatment of Data

The data gathered will be analyzed using descriptive statistical tools. Frequency and Percentage Distribution will be used to organize the respondent profile, distinguishing between Admin and Client users. To directly address the third statement of the problem regarding software quality, the Weighted Mean will be computed to determine the respondents' central assessment of the system's ISO/IEC 25010 characteristics. Finally, Standard Deviation will be utilized to measure the consistency of the responses, indicating the degree of consensus among users regarding the system's effectiveness in resolving the identified operational inefficiencies

E. Software Development Life Cycle :

The development of the PrintSmart system will follow the **Agile Iterative Development** methodology. This approach

is selected to directly address the first statement of the study's problem, the operational inefficiencies in the current manual, Facebook-based process, by allowing for continuous feedback and refinement throughout the build process.

Sprint 0: Planning and Requirements Analysis
This initial phase addresses the first objective of the study: identifying the specific inefficiencies in order tracking, file management, and payment verification. Through interviews with Printagram staff, the researchers will map the current manual workflow to ensure the new unified web platform is designed to resolve these specific bottlenecks.

Sprint 1 & 2: System Design and Core Development
These sprints focus on the second objective: the design and development of the **Client-Side** and **Admin-Side** modules. During this phase, the researchers will construct the user-friendly interface featuring the e-commerce catalog and shopping cart, as well as the secure payment gateway integration. Simultaneously, the centralized admin dashboard will be developed to handle real-time order tracking. This iterative construction ensures that every feature developed directly corresponds to the functional requirements needed to facilitate seamless online transactions.

Sprint 3 & 4: Testing, Deployment, and Evaluation
The final sprints are dedicated to the third objective: evaluating the system based on ISO/IEC 25010 standards. After deploying the stable system, it will undergo testing by client and admin users. The feedback gathered during this phase will determine if the system has successfully achieved the necessary **Functional Suitability, Usability, Security, Reliability, and Performance Efficiency** to replace the previous manual operations and enable the business to reach a wider market.

F. Ethical Considerations

The research and development of the PrintSmart system strictly adhere to ethical standards and the principles of Confidentiality, Integrity, and Availability (CIA) to safeguard participant data.

Informed Consent and Voluntary Participation
Prior to data collection, all participants, including Printagram staff and selected clients, will be fully informed about the purpose of the research, the procedures to be conducted, potential risks (which are minimal), and the benefits of participation. They will also be made aware of their right to refuse participation or withdraw from the study at any point without any negative consequences. To formalize this agreement, signed consent forms will be obtained from all participants, or, in the case of online surveys, explicit consent will be indicated through checked acknowledgment boxes before participation.

Confidentiality (Privacy and Anonymity)
To ensure confidentiality, the privacy of all participants will be protected by omitting personal identifiers and assigning unique codes to each respondent. Only aggregated data, such as frequency tables, percentages, or averages, will be presented in the final research report, ensuring that individual responses cannot be traced back to specific participants.

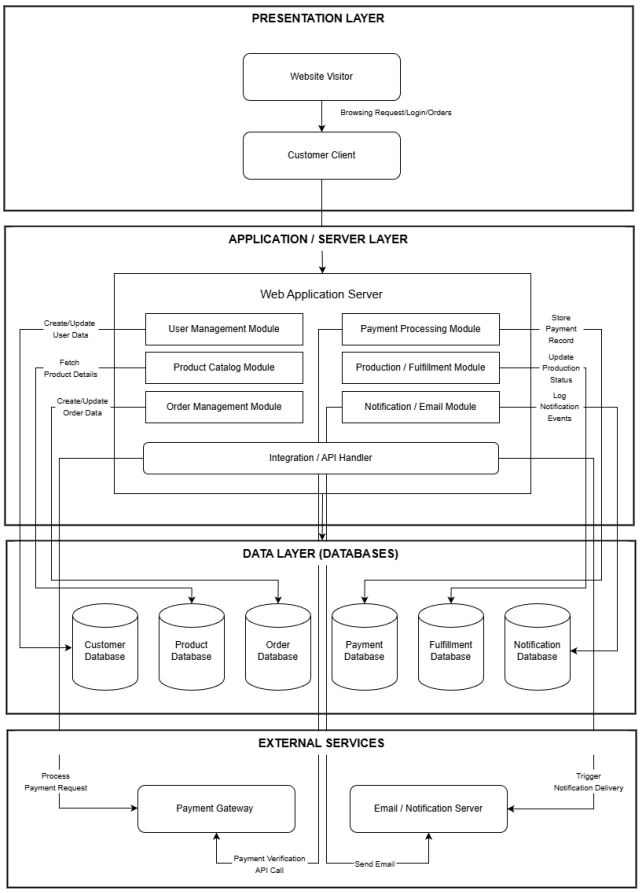
Integrity (Data Protection and Accuracy)
Integrity is upheld by complying with the principles of the Philippine Data Privacy Act, ensuring that participants' personal information is safeguarded against unauthorized alteration or misuse. The researchers commit to handling electronic and physical data with care to prevent unauthorized access, ensuring the data remains accurate and trustworthy throughout the study.

Availability (Secure Access)
To maintain security while ensuring data availability for valid research purposes, all research data will be stored securely in password-protected files accessible only to the researchers and their adviser. This restriction ensures that the data is available for analysis and verification by authorized personnel but remains inaccessible to unauthorized third parties.

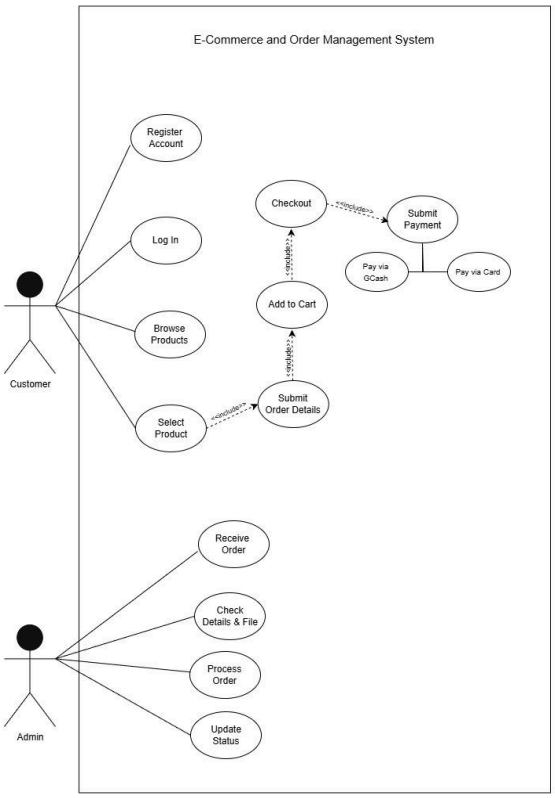
G. System Architecture

This section presents the architectural design and logical structure of the PrintSmart system. It details the hierarchical layers of the application, the functional interactions of users, the flow of data across processes, and the database schema used to store information.

1. System Architecture (Design by Level of Application)
The System Architecture Diagram illustrates the multi-layer structure of the PrintSmart system, organized into the Presentation Layer, Application/Server Layer, Data Layer, and External Services Layer. The diagram highlights how the user interface interacts with the web application modules, such as User Management, Payment Processing, and Production/Fulfillment, while accessing the various databases. Furthermore, it demonstrates the integration with external services, specifically the Payment Gateway for transaction verification and the Email Server for automated notifications.



Unified Modeling Language (UML)

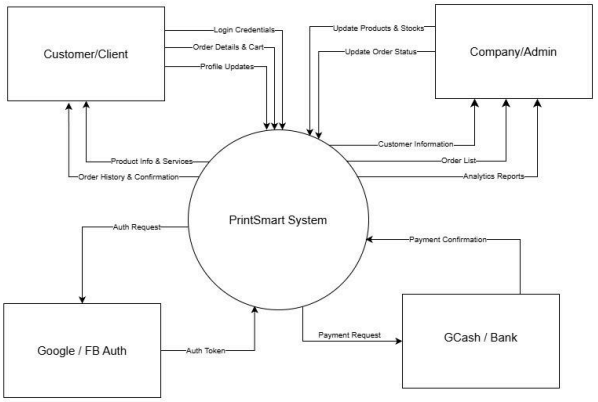


The Use Case Diagram illustrates the primary interactions between the system and its external users, identifying two main actors: the Customer and the Admin. The diagram outlines specific functionalities, showing that customers can browse products, place orders, upload design files, and make payments. Conversely, administrators are authorized to manage products, process orders, monitor payment transactions, and update order statuses. This visualization provides a high-level overview of the system's functional requirements and user roles.

Data Flow Diagrams (DFD)

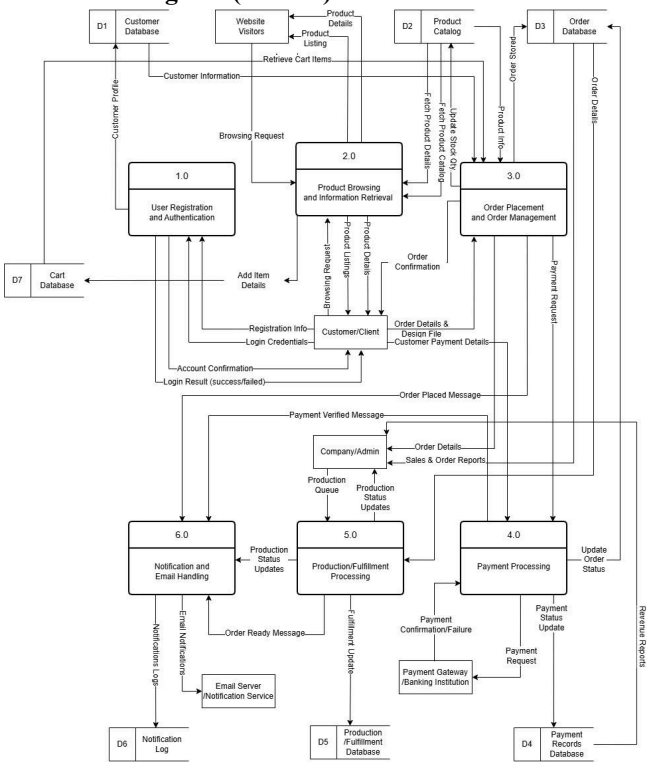
To illustrate the movement of data within the system, two levels of Data Flow Diagrams were designed.

Context Diagram (Level 0):



This diagram presents PrintSmart as a single process, highlighting how it exchanges data with external entities including the Customer, Admin, Payment Gateway, and Email/Notification Server. It defines the system boundaries and the primary inputs and outputs, such as order details, payment confirmations, and system notifications.

Data Flow Diagram (Level 1):



Expanding on the context diagram, the Level 1 DFD breaks down the system into major subprocesses including user authentication, product browsing, order processing, and payment handling. It details how data moves between these processes, data stores, and actors, illustrating the internal transformation of inputs into system outputs.

H. Time line

The development of the PrintSmart system followed the Agile Iterative Development life cycle, divided into five distinct sprints (Sprint 0 to Sprint 4). This approach allowed for continuous refinement of features based on user feedback and operational requirements. The specific activities and expected outputs for each phase are detailed in Table I.

Phase	Date	Activity
Planning	Oct. 21–24, 2025	Establishing key stakeholders, defining the project scope, and identifying the core issues and objectives for the research.
Analysis	Oct. 27–31, 2025	Conducting interviews, observations, and gathering relevant data to inform system development and decision-making.
Designing	Nov. 3–7, 2025	Drafting the system design, preparing required equipment, and planning the database structure to align with project goals.
Development	Nov. 10, 2025 – Jan. 3, 2026	Creating an initial prototype of the system and device, showcasing a working model of 40–60% of the final product.
Review and Validation	Jan. 5–9, 2026	Revising project documentation and preparing necessary materials for review and approval in preparation for the oral defense.
Testing	Jan. 12–17, 2026	Conducting comprehensive system testing and validation during the oral defense evaluation to ensure functionality and effectiveness.
Deployment	Jan. 19–23, 2026	Filing necessary UREC documents, validating research instruments, and obtaining ethical clearance for the study.

TABLE I. PROJECT DEVELOPMENT TIMELINE

Acknowledgment

References

[1] M. Joshi, Intro to E-commerce and Social Commerce. Educohack Press, 2025.

[2] R. W. Attar, A. Almusharraf, A. Alfawaz, and N. Hajli, "New trends in e-commerce research: Linking social commerce and sharing commerce: A systematic literature review," Sustainability, vol. 14, no. 23, p. 16024, 2022.

[3] A. P. Esteban, Web Engineering and E-commerce: Bridging Technology and Business in the Philippines. Nueva Ecija Univ. of Science and Technology, 2023.

[4] B. Barnabas and M. Idowu, "Digital synergy: Advancing e-commerce through web engineering in the Philippines," 2025. [Online]. Available: ResearchGate.

[5] E. Blancaflor and J. Jocson, "LocaleCommerce: Empowering local markets with NopCommerce, an open-source e-commerce platform," in ACM Digital Library, 2024. doi: 10.1145/3695652.3695655.

[6] D. Novitasari and A. H. Wijaya, "Web-based ordering and inventory information system for online printing services with TAM method," Tech-E: J. Tech-E, vol. 5, no. 1, 2021.

[7] MCC Institute, "MCC-IT research journal 2022-2023," 2023. [Online]. Available: https://mcc.edu.ph/academic-research/download/MCC-IT%20Research%20Journal%202022-2023.pdf

[8] R. Sidik, N. Rosmawati, and S. Mauluddin, "Designing a customized order management system prototype for printing products and services: From concept to development," in 2023 Int. Conf. Informat. Eng., Sci. & Technol. (INCITEST), pp. 1-7, IEEE, 2023.

[9] J. L. Alcorin, R. J. Climaco, E. L. Jabat, and J. A. Simbaco, "A web-based inventory tracking with customer ordering, point-of-sale system and AI integration for clothing and printing shops in San Jose del Monte," Capstone project proposal, STI College San Jose del Monte, April 2025.

[10] R. Thakor, "Common challenges in payment gateway integration and how to solve them," The One Technologies, 2024. [Online]. Available: https://theonetechnologies.com/blog/post/common-challenges-in-payment-gateway-integration-and-how-to-solve-them

[11] A. C. Dewi, E. I. H. Ujianto, and R. Rianto, "Electronic payment threats and security: A systematic literature review," Jurnal Ilmiah J-JANAPATI, vol. 1, no. 1, pp. 1-10, 2024.

[12] Antom, "Young and local: How Filipino consumers are embracing digital payments and e-commerce," Antom Knowledge, Sept. 8, 2025. [Online]. Available: https://www.antom.com/knowledge/young-and-local-how-filipino-consumers-are-embracing-digital-payments-and-e-commerce/

[13] T. J. C. Piad, "Data demand, GCash popularity lift Globe income by 6%," Philippine Daily Inquirer, Nov. 9, 2024.

[14] H. Siagian, Z. Tarigan, and R. Basuki, "The effect of perceived security, perceived ease of use, and perceived usefulness on consumer behavioral intention through trust in digital payment platform," Int. J. Data Netw. Sci., vol. 6, no. 3, pp. 861-874, 2022.

[15] A. O. Ali, "The payments security and convenience on consumer of mobile payments in Baguio City, Philippines," Eur. J. Manage. Marketing Stud., vol. 9, no. 3, 2024.

[16] L. Martins, R. Silva, and P. Rodrigues, "User-centric principles and usability enhancement in e-commerce platforms," J. Digit. Commer. Stud., vol. 15, no. 2, pp. 44-59, 2021.

[17] H. Yoon and S. Lim, "Understanding user adoption of e-commerce platforms using the Technology Acceptance Model (TAM)," Int. J. E-Bus. Res., vol. 19, no. 1, pp. 22-37, 2023.

[18] A. Rahman and L. Wong, "User interface satisfaction and acceptance of e-commerce applications among Southeast Asian consumers," Asia-Pacific J. Human-Comput. Interact., vol. 9, no. 2, pp. 61-79, 2022.

[19] J. P. Soriano and M. R. Aquino, "User-experience design in online business systems among Filipino micro-enterprises," Philippine J. E-Commer. Innov., vol. 4, no. 1, pp. 11-28, 2021.

[20] N. H. Mullick and N. Rafique, "A study on moderating effect of customization on antecedents of customer satisfaction in online apparel shopping," J. Informat. Educ. Res., vol. 3, no. 2, 2023.

[21] Z. Li, H. Zhang, and J. Xue, "E-commerce cart recommendation effects: A field experiment on entertainment products," 2023. [Online]. Available: https://justc.ustc.edu.cn/article/pdf/preview/JUSTC-2022-0130.pdf

[22] IndexBox, "Asia-Pacific's printing press market: Reaching 2.3 M units and $4.8 B by 2035," 2024. [Online]. Available: https://www.indexbox.io/blog/printing-press-asia-pacific-market-overview-2024-1/

[23] Y. K. Cho, C. L. Sutton, and N. Taskin, "Positive relationships between service performance and social media use in internet retailing: Does information symmetry matter?" Contemp. Manage. Res., vol. 19, no. 3, pp. 207-233, 2023.